

Air Quality Prediction Using Environmental Sensor Data

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Table of Contents

Abstract	4
1. Business Understanding.....	5
1.1. Background.....	5
1.2. Problem Statement	5
1.3. Objectives.....	7
1.4. Success Criteria	8
2.1. Dataset source and context	8
2.2. Dataset Structure and Variables	9
2.3. Initial Data Exploration.....	10
2.4. Exploratory Data Analysis (EDA).....	11
3. Data Preparation.....	13
3.1. Missing Values and Data Cleaning.....	13
3.2. Feature Engineering and Transformation	14
3.3. Variable Selection and Data Transformation	15
4. Modeling	16
4.1 Regression Models	16
4.1.1. Baseline Model: Sensor-Based Regression.....	16
4.1.2. Refined Model: Regression Without Absolute Humidity (AH)	18
4.1.3. Extended Model: Regression with Hour	20
4.1.4. Model 4: Extended with Weekday Effects.....	25
4.2 Model Diagnostics.....	28
4.3 Additional Time Insights	29
4.4 Out-of-Sample Forecasting	30
5. Evaluation	32
5.1 Model Performance Metrics.....	32
5.2 Performance Summary and Comparative Interpretation.....	33
5.3 Reliability of Results	34

5.4 Final Model Selection	36
5.5. Cross Validation and Model Generalization	37
6. Deployment & Reflection.....	38
6.1 Practical Applications	38
6.2 Limitations.....	38
6.3 Future Improvements	39
6.4 Final Insight	39
References.....	41

Abstract

Air pollution is considered one of the most important environmental problems, as it affects many cities around the world. According to the World Health Organization (2021), air pollution represents a major global health risk, especially in urban areas. Among different air pollutants, carbon monoxide (CO) is particularly harmful because it can significantly affect human health. This pollutant is mainly produced by traffic and industrial activities.

Understanding the environmental factors related to CO is essential to improve air quality monitoring and support environmental decision-making processes.

Additionally, this study analyzes environmental sensor data to explore how atmospheric conditions are related to CO concentrations. The Air Quality dataset was the base chosen to apply statistical modeling techniques to estimate CO levels based on environmental variables and sensor measurements. The analysis follows the CRISP-DM framework, so it is composed by stages such as data understanding, data preparation, modeling, evaluation, and reflection.

Moreover, a multiple linear regression model was the method selected to predict carbon monoxide concentration using variables such as sensor readings, temperature, humidity, and other atmospheric indicator as predictors. The results exhibits that the sensor variable PT08.S1(CO) has a strong positive relationship with CO levels. In addition, temperature and relative humidity have statistically significant effects but the multicollinearity is an issue to be addressed. According to the results the final model is able to explain almost 88.6% of the variation in CO levels, showing a strong explanatory capacity.

Finally, the results obtained highlight the significance of environmental monitoring systems and show how statistical models can be used to analyze pollution patterns and support predictive environmental analysis.

1. Business Understanding

1.1. Background

Currently, air pollution is considered one of the most significant environmental challenges worldwide, particularly in urban areas where pollution levels are influenced by traffic congestion, industrial emissions, and atmospheric conditions that limit pollutant dispersion (World Health Organization, 2021).

Carbon monoxide (CO) is especially dangerous due to its characteristics, as it is a colorless and odorless gas that interferes with oxygen transport in the bloodstream, potentially leading to serious health consequences.

Environmental agencies rely on sensor networks to monitor pollution levels. However, certified analyzers are expensive and difficult to deploy at large scale. For this reason, recent research has explored the use of low-cost chemical sensors combined with statistical and machine learning methods to estimate pollutant concentrations more efficiently (De Vito et al., 2008).

Nowadays, predictive models based on environmental variables can support the estimation of pollution levels and contribute to improving monitoring strategies and decision-making processes in urban environments.

1.2. Problem Statement

In most environmental monitoring systems, pollutants such as carbon monoxide (CO) require specialized equipment. According to De Vito et al. (2008), while these certified analyzers provide accurate measurements, they are expensive, require continuous maintenance, and cannot

be easily deployed at large scale. Due to this, there is increasing interest in finding alternative ways to estimate pollution levels using more accessible data.

This study was conducted based on a key question:

Can we accurately estimate carbon monoxide (CO) levels using indirect information such as sensors, environmental conditions, and time-based patterns?

One of the difficulties is how data from sensor works. Any sensors like PT08.S1(CO) do not measure CO directly from the environment, otherwise they take signals influenced by different gases and environmental conditions. Doing this the device is able to establish a relationship between its readings and the actual CO concentration.

Additionally, environmental factors such as temperature and humidity should be considered due to, they can affect both pollution and sensors behave. Through these variables it is possible to introduce an important variability that should be considered when building predictive models.

Moreover, time is another relevant aspect because air pollution varies throughout the day, since the increases patterns are linked to human activities, such as traffic during peak hours, as well as changes in weather conditions. Including features like the hour of the day or the day of the week improve the ability of the model to capture these patterns.

Taking these factors as reference, this project tries to solve the following problem:

How well sensor data, environmental variables and temporal information can predict real pollution levels.

Finally, this study has real world relevance, because it can support the development of more affordable and scalable air quality monitoring systems.

1.3.Objectives

This project has as main objective evaluate whether carbon monoxide (CO) concentrations can be accurately predicted using indirect data sources such as sensor measurements, environmental conditions and temporal patterns.

To achieve this objective, this project has been structured based on the following specific objectives:

Primary, to develop EDA in order to understand the dataset chosen, identifying key variables, data types, and other potential issues that can appear such as missing values or data inconsistencies. This stage is critical because helps build a good foundation for the analysis.

Secondly, preprocess the data to ensure data quality and usability. This preprocess step includes handling missing values, selecting pertinent variables, and preparing the data for modeling.

Thirdly, to develop a predictive model that estimates CO concentrations based on readings such as PT08.S1(CO), environmental conditions like temperature, humidity and absolute humidity, and temporal features for example time and date.

Fourthly, to evaluate model performance according to statistical metrics and to assess how well the chosen variables can explain variations in CO levels.

Finally, analyzing the contribution of different variable types for instance sensor based, environmental, temporal, to understand what the most important factor is to predict air pollution.

In general, these objectives help to provide a technical evaluation of the model and also offers an idea to understand how different data sources can be used to estimate air quality in a practical and scalable way.

1.4. Success Criteria

The success of this project is evaluated based on the combination of statistical performance, methodological rigor, and practical interpretability. The regression model should demonstrate strong explanatory power, which is reflected by a high R^2 value, which indicate that an important proportion of the variation in CO concentrations can be explained by the selected predictors. Furthermore, individual coefficients should be statistically significant and consistent with theoretical expectations. Model diagnostics, residual analysis and multicollinearity assessment using Variance Inflation Factor (VIF) must confirm that the required assumptions of linear regression are satisfied and ensure the reliability and stability of the results.

Moreover, the model should provide insights into how different types of variables contribute to predicting air pollution levels. In this case, a successful outcome will show that information obtained from sensor could be a strongest predictive signal, but the inclusion of other variables enhances the ability of the model to capture real world pollution patterns, this perspective offers a more realistic representation of environmental dynamics and increase the explanatory power.

Finally, this study can be considered successful if the results are interpretable, reproducible, and applicable to real world. The model must show that the inclusion of other variables plus accessible data sources can be used to estimate pollutant concentrations with accuracy.

2. Data Understanding

2.1. Dataset source and context

The dataset chosen for this project is he Air Quality dataset from the UCI Machine Learning Repository, this dataset was originated from a scientific experiment conducted by De

Vita et al. (2008), all the observations included into the dataset were collected in an urban area in Italy and the study was developed between March 2004 and February 2005, each observation was recorded per hour, obtaining 9,357 observations over the study period.

The experiment was developed to evaluate the performance of low-cost metal oxide chemical sensors in detecting air pollutants. The sensors were located next to certified reference analyzers that provided accurate measurements of pollutants such as carbon monoxide (CO), nitrogen dioxide (NO₂), nitrogen oxides (NO_x), and benzene. Additionally, meteorological variables such as temperature, relative humidity and absolute humidity were registered. The experimental set up makes the dataset valuable for develop a statistical modeling due to it allows the comparison between indirect sensor responses and real pollutant concentrations, enabling the development of predictive models for air quality estimation.

2.2.Dataset Structure and Variables

The dataset is composed by 9,357 observations and 15 variables, meeting the minimum requirements for the assignment, the variables in the dataset represent different aspects of environmental monitoring and can be classified in three classes like sensor, environmental, and temporal components.

The target variable, CO(GT) is the real concentration of Carbon Monoxide and is the dependent variable of the analysis, moreover, sensor variables, particularly PT08.S1(CO), is the responses of the sensors and has a strong relationship with CO levels. In addition, sensor variables is the measure related to other gases and contribute a representation of air quality conditions.

Furthermore, environmental variables describe atmospheric conditions that may influence pollutant behavior and sensor responses. Finally, temporal features were derived from the dataset

by extracting the hour of the day from the time variable and the day of the week from the date variable. The engineered features were incorporated to capture temporal pattern that can be associated with pollution levels, and the variations due to daily human activity and environmental cycles.

2.3.Initial Data Exploration

The initial exploration of the dataset was focused on review the structure and content of it, this step was developed using standard data inspection techniques. For example, the starting point was displaying the first observations, reviewing variable types, and verifying the dimension of the data. This stage was crucial to check if the dataset met the assignment requirements and to obtain a quick understanding of how the variables were organized and recorded.

In this stage was possible to do the identification of inconsistencies and data quality issues. One of the findings was that missing values were not labeled as null instead of that the missing values were encoded using the value of -200, this represents sensor malfunction or invalid readings. In this type of study is important to treat these values as missing values because if these values are treated as valid observations, it would introduce bias into the analyses and model estimation.

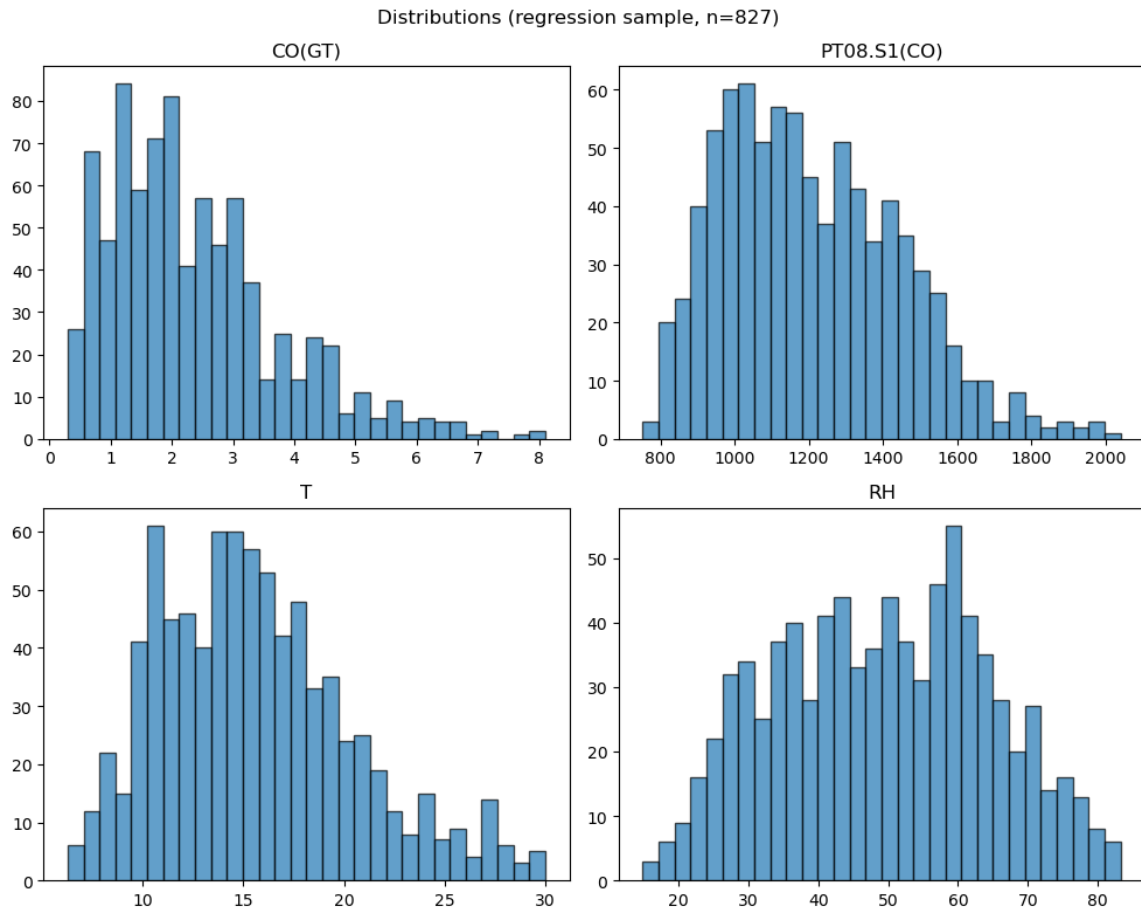
Moreover, the statistics were examined to identify the central tendency and dispersion of the variables. Since measures such as mean, standard deviation, and range provided insights related to predictors and target variables. This review allowed identify anomalies, such as large or small values, additionally, the data types were reviewed to ensure that numerical variables were correctly formatted for analysis and that the temporal variables could be transformed into features such as hour and weekday.

Overall, this stage provided an understanding of the dataset's structure and reveal issues such as the presence of nonstandard missing values.

2.4. Exploratory Data Analysis (EDA)

The exploratory data analysis (EDA) was conducted to better understand the distribution of variables, identify patterns, and examine relationships between predictors and the target variable. This stage is responsible for guiding the modeling strategy, due to this it provides insights related to the structure of the data as skewness, outliers, and multicollinearity.

Figure 1 Distributions (Regression sample, n= 827)

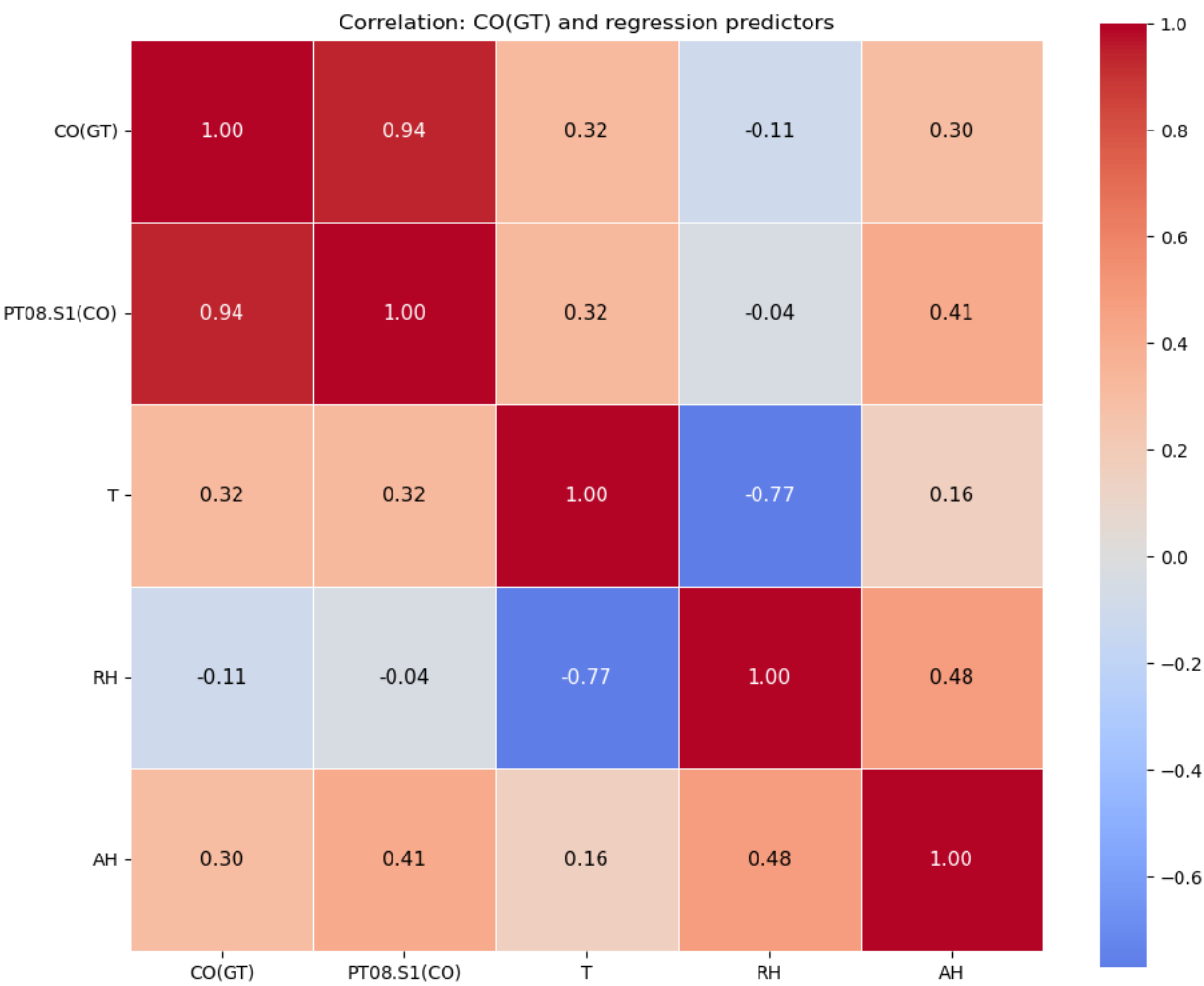


The first step was to analyze the distribution of key variables, including carbon monoxide concentration (CO(GT)), sensor measurements, and environmental variables such as temperature

and relative humidity. The plot reveals that several variables do not follow a normal distribution, specifically, CO concentrations and sensor readings presents skewness and variability, something that is common in environmental datasets due to fluctuations in atmospheric conditions and human activity. On the other hand, temperature exhibits a more symmetric distribution, and relative humidity shows a wider dispersion. The findings suggest that linear regression is

Figure 2 Correlation: CO(GT) and Regression predictors

appropriate and the data contains a variability that reflects real world environmental conditions.



Secondly, a correlation analysis was developed to quantify the strength and direction of relationship among variables. The correlation matrix presents a very strong correlation (0.94) between CO and PT08.S1(CO), confirming the important role of sensor measurements as the primary predictor. Additionally, there are moderate correlations between CO and environmental variables, this indicates that natural conditions contribute to explain pollution levels but are not dominant drivers.

An important finding of this matrix was the presence of strong correlations among meteorological variables like temperature, relative humidity, and absolute humidity. This correlation indicates the presence of multicollinearity that can affect regression models inflating standard errors and reducing the interpretability of coefficients. The identification of this issue at this stage was crucial to adjust the model specification during the modeling phase.

Overall, EDA provided important insights into the structure and behavior of the dataset. It was possible to confirm the strong predictive potential of sensor measurements, and to identify the secondary role of environmental variables. Additionally, this stage revealed the existence of multicollinearity among meteorological features. Furthermore, the variability and distribution patterns suggest the importance of data processing and model diagnosis.

3. Data Preparation

3.1.Missing Values and Data Cleaning

Identifying and handle missing into the dataset is an important stage of the study, during the exploratory phase, it was observed that missing values, missing or invalid sensor readings were not coded as null values instead of that they were represented using the value -200. This labeled is normal in environmental sensor datasets, this value can indicate malfunction or wrong

measurements, as it was mentioned earlier these values cannot be treated as valid observation because it would introduce bias into the analysis and affect the estimation of statistical relationships.

This issue was addressed replacing all instances of -200 with NaN values as is usual in standard data cleaning procedures. Following the transformation process, all the rows containing missing values were removed using `dropna()` method. This step ensured that the data used for modeling contained observed and reliable data into the selected variables, this approach reduced the size of CO observation to 827 note that the dataset has additional observations as NHM, C6H6 and NO2.

Although alternative imputation methods could have been considered, such as mean imputation or interpolation, the decision taken to remove incomplete observations is justified by the nature of missing values, and it was mentioned earlier these values represent sensor errors, imputing them can introduce fake patterns and compromise the validity of the model.

3.2.Feature Engineering and Transformation

To improve the predictive capability of the model and better capture real world pollution patterns, additional features were engineered from the original dataset, the focus was on temporal information. The raw date and time variables were transformed into more analytical components. In specific, the hour of the day was obtained from the time variables and allows the model to capture intraday variations in pollution levels, such as peak traffic hours and reduced nighttime activity.

Additionally, the day of the week was obtained from the date variable to account for weekly patterns in human activity and environmental conditions. It is common that air pollution

levels differ between weekdays and weekends due to variations in traffic, industrial operations, and urban activity. By incorporating this feature, the model is able to capture temporal patterns.

Incorporating the categorical nature of the day of the week variable into the regression model, one-hot encoding was applied. This transformation converts each category into a binary indicator variable, allowing the model to estimate separate effects for each day without imposing an artificial ordinal structure. The inclusion of these engineered temporal features allows the model to incorporate time-based behavioral patterns.

3.3.Variable Selection and Data Transformation

After data cleaning and feature engineering, the selection of variables was performed to ensure that the model included relevant predictors to avoid redundancy and instability. The primary predictor selected was the sensor variable PT08.S1(CO), which showed a very strong correlation with the target variable CO(GT) during the exploratory analysis. This variable has a main role of predictive power in the model, due to reflect the direct relationship between sensor response and pollutant concentration.

Additionally, environmental predictors such as temperature (T) and relative humidity (RH) were included to capture atmospheric conditions that influence pollutant behavior. Although these variables present weaker correlations with CO levels compared to the sensor measurement, they provide important information and improve the explanatory power of the model. However, due to the presence of multicollinearity into environmental variables certain variables were exclude during the modeling phase to ensure model stability and interpretability.

Furthermore, all variables were verified in order to ensure that all the variables have the appropriate format for regression analysis, and it was not necessary to apply additional scaling or normalization. Overall, the data preparation gives as a result a clean, consistent and well-

structured dataset which integrates sensor measurements, environmental conditions, and temporal features, providing a solid foundation for the modeling phase.

4. Modeling

4.1 Regression Models

This section presents a series of regression models developed to explain the variation in CO(GT). The modeling process follows a step-by-step approach, starting with a baseline model and progressively refining it based on statistical significance, multicollinearity, and the inclusion of time-related variables.

The goal is not only to improve model performance, but also to better understand which factors drive CO levels. Each model builds on the previous one, allowing for a clearer interpretation of how sensor, environmental, and temporal variables contribute to the results.

4.1.1. Baseline Model: Sensor-Based Regression

The first regression model was built using the sensor variable PT08.S1(CO) together with the environmental variables T, RH, and AH to predict CO(GT). The goal of this model is to serve as a baseline and to understand how much of the variation in CO it explains before making any adjustments to the model.

As shown in Figure 3, the model has an R^2 of 0.886, which means the model captures most of the variation in CO levels. However, when looking more closely at the results, it becomes clear that this performance is mostly driven by one variable. The coefficient for PT08.S1(CO) is positive ($\beta = 0.0057$) and highly significant ($p < 0.001$), showing a strong relationship between the sensor readings and CO.

Figure 3 OLS regression results for CO(GT) using sensor and environmental variables

OLS Regression Results

Dep. Variable:	CO(GT)	R-squared:	0.886
Model:	OLS	Adj. R-squared:	0.885
Method:	Least Squares	F-statistic:	1595.
Date:	Sat, 14 Mar 2026	Prob (F-statistic):	0.00
Time:	23:48:35	Log-Likelihood:	-559.28
No. Observations:	827	AIC:	1129.
Df Residuals:	822	BIC:	1152.
Df Model:	4		
Covariance Type:	nonrobust		

	coef	std err	t	P> t	[0.025	0.975]
const	-3.5573	0.286	-12.431	0.000	-4.119	-2.996
PT08.S1(CO)	0.0057	7.86e-05	72.032	0.000	0.006	0.006
T	-0.0137	0.016	-0.880	0.379	-0.044	0.017
RH	-0.0070	0.006	-1.275	0.203	-0.018	0.004
AH	-0.4423	0.309	-1.432	0.153	-1.049	0.164

Omnibus:	29.759	Durbin-Watson:	0.513
Prob(Omnibus):	0.000	Jarque-Bera (JB):	57.245
Skew:	0.227	Prob(JB):	3.71e-13
Kurtosis:	4.206	Cond. No.	2.99e+04

On the other hand, the environmental variables T, RH, and AH are not statistically significant, since their p-values are above 0.05. This suggests that once the sensor is included, these variables do not add much new information to the model. Their contribution is small and overlaps with what the sensor is already capturing.

This idea is supported by the results of the multicollinearity analysis shown in Figure 4. The VIF values indicate that T, RH, and AH are highly correlated. In particular, AH exceeds the usual threshold, suggesting it introduces redundancy and makes the model less stable.

Figure 4 Variance Inflation Factor (VIF) results for model predictors

	variable	VIF
0	const	297.283815
1	PT08.S1(CO)	1.309665
2	T	20.390888
3	RH	25.716111
4	AH	11.023311

In addition, the condition number reported in Figure 4 is relatively high, which also points to multicollinearity in the model. Together, Figures 1 and 2 show that some variables are overlapping and not improving the model.

Overall, the results suggest that most of the model's performance comes from the sensor variable. PT08.S1(CO) captures most of the variation in CO(GT), while the environmental variables add limited value in this version of the model. Based on this, the next step is to simplify the model by removing redundant variables and improving its stability.

4.1.2. Refined Model: Regression Without Absolute Humidity (AH)

The second model was built by removing the variable Absolute Humidity (AH). This adjustment was made based on the baseline model results, in which AH showed high multicollinearity and was not statistically significant.

As shown in **Figure 5**, the model maintains an R^2 of 0.886, which is the same as in the full model. This shows that removing AH does not reduce the model's ability to explain CO(GT). In other words, AH did not add meaningful information to the model.

Figure 5 OLS regression results for CO(GT) without Absolute Humidity (AH)

OLS Regression Results						
=====						
Dep. Variable:	CO(GT)	R-squared:	0.886			
Model:	OLS	Adj. R-squared:	0.885			
Method:	Least Squares	F-statistic:	2124.			
Date:	Sat, 14 Mar 2026	Prob (F-statistic):	0.00			
Time:	23:48:35	Log-Likelihood:	-560.31			
No. Observations:	827	AIC:	1129.			
Df Residuals:	823	BIC:	1147.			
Df Model:	3					
Covariance Type:	nonrobust					
=====						
	coef	std err	t	P> t	[0.025	0.975]

const	-3.2174	0.160	-20.118	0.000	-3.531	-2.903
PT08.S1(CO)	0.0056	7.75e-05	72.765	0.000	0.005	0.006
T	-0.0342	0.006	-5.611	0.000	-0.046	-0.022
RH	-0.0145	0.002	-7.956	0.000	-0.018	-0.011
=====						
Omnibus:	29.754	Durbin-Watson:	0.514			
Prob(Omnibus):	0.000	Jarque-Bera (JB):	57.316			
Skew:	0.227	Prob(JB):	3.58e-13			
Kurtosis:	4.207	Cond. No.	1.19e+04			
=====						

At the same time, the results become more consistent. The sensor variable PT08.S1(CO) remains positive and highly significant ($p < 0.001$), confirming that it continues to be the main driver of the model. Unlike the previous specification, the environmental variables T and RH are now statistically significant, with p-values below 0.05. This suggests that, once the redundant variable is removed, their individual contributions become clearer.

This improvement is supported by the results of the multicollinearity analysis shown in **Figure 6**. The VIF values for all predictors are now below 5, indicating that multicollinearity is no longer an issue. Compared to the previous model, where VIF values were above 10, this represents a clear improvement in model stability.

Figure 6 Variance Inflation Factor (VIF) results after removing AH

	variable	VIF
0	const	92.731368
1	PT08.S1(CO)	1.273305
2	T	3.124440
3	RH	2.798937

In addition, the condition number is lower than in the baseline model, suggesting that the relationships between variables are now more stable and less affected by overlap.

Overall, this refined model is easy to understand and works better. It maintains the same level of explanation for CO(GT) but avoids including variables that were not contributing much to the model. By removing AH, the model becomes simpler, and this makes the model more reliable and easier to interpret.

This process can be considered a form of model refinement, where variables are adjusted based on statistical significance and multicollinearity diagnostics to improve model stability and interpretability.

4.1.3. Extended Model: Regression with Hour

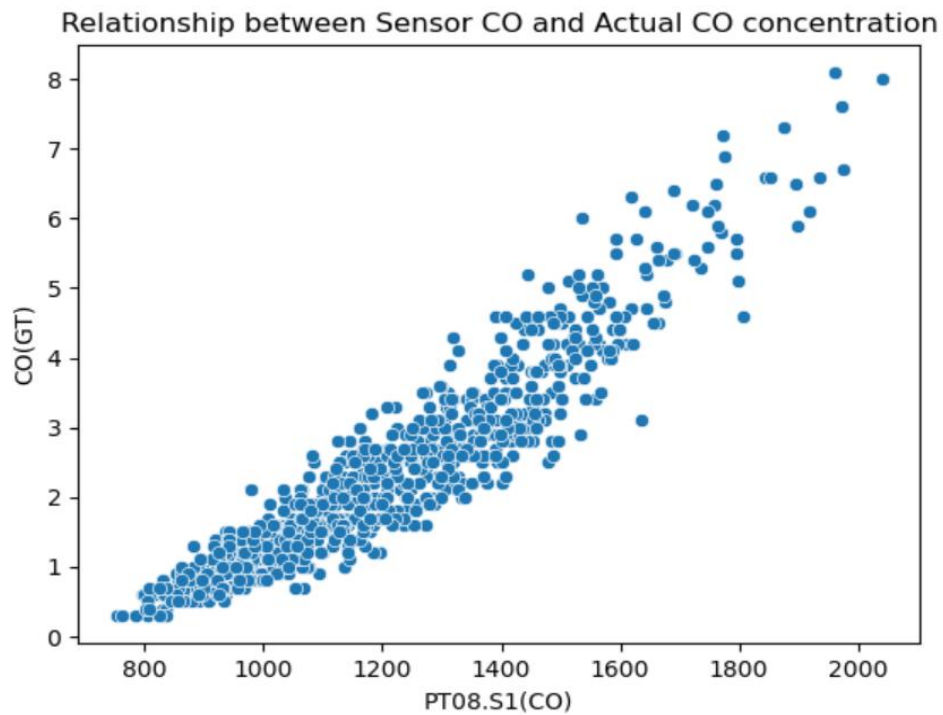
To further improve the model, the variable hour of the day was introduced. The goal is to capture time-related patterns in CO(GT) that are not explained by the sensor and environmental variables alone.

Even though the previous model showed good performance, it did not account for how CO levels may change throughout the day. By adding the hour variable, the model can capture common daily patterns in air pollution data.

Before including this variable in the regression, the relationship between the predictors and CO(GT) was explored using scatter plots. As shown in **Figure 7**, there is a strong positive

relationship between the sensor variable PT08.S1(CO) and CO(GT), showing that the sensor is still the main driver of the model.

Figure 7 Relationship between PT08.S1(CO) and CO(GT)



In contrast, the relationships for T and RH (**Figures 8 and 9**) appear weaker and more dispersed, which supports their smaller effect in the regression.

Figure 8 Temperature (T) vs CO(GT)

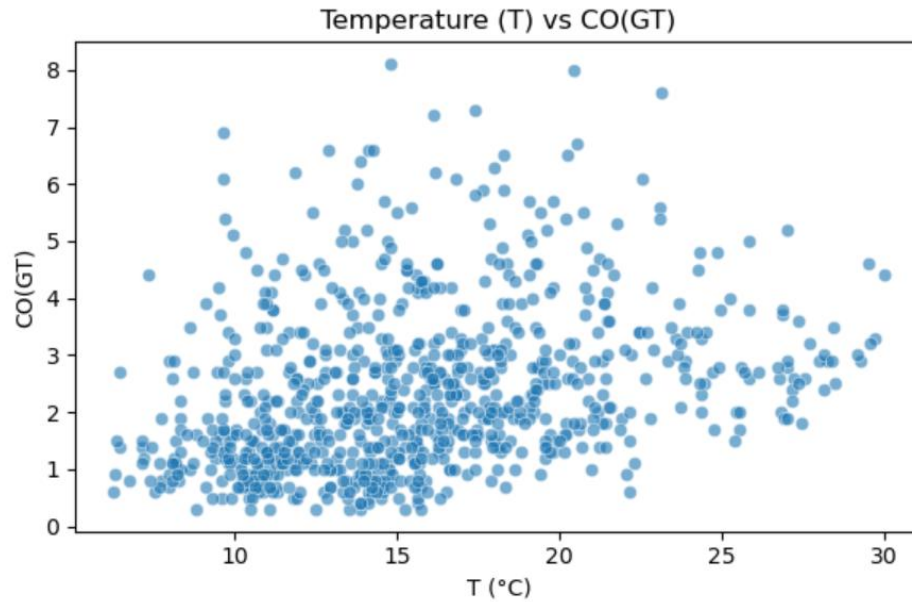
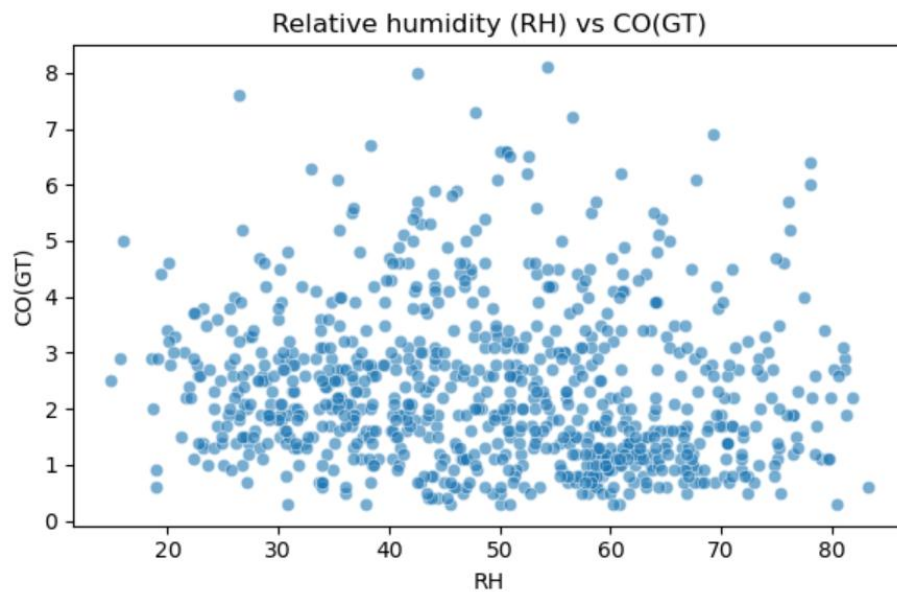


Figure 9 Relative Humidity (RH) vs CO(GT)



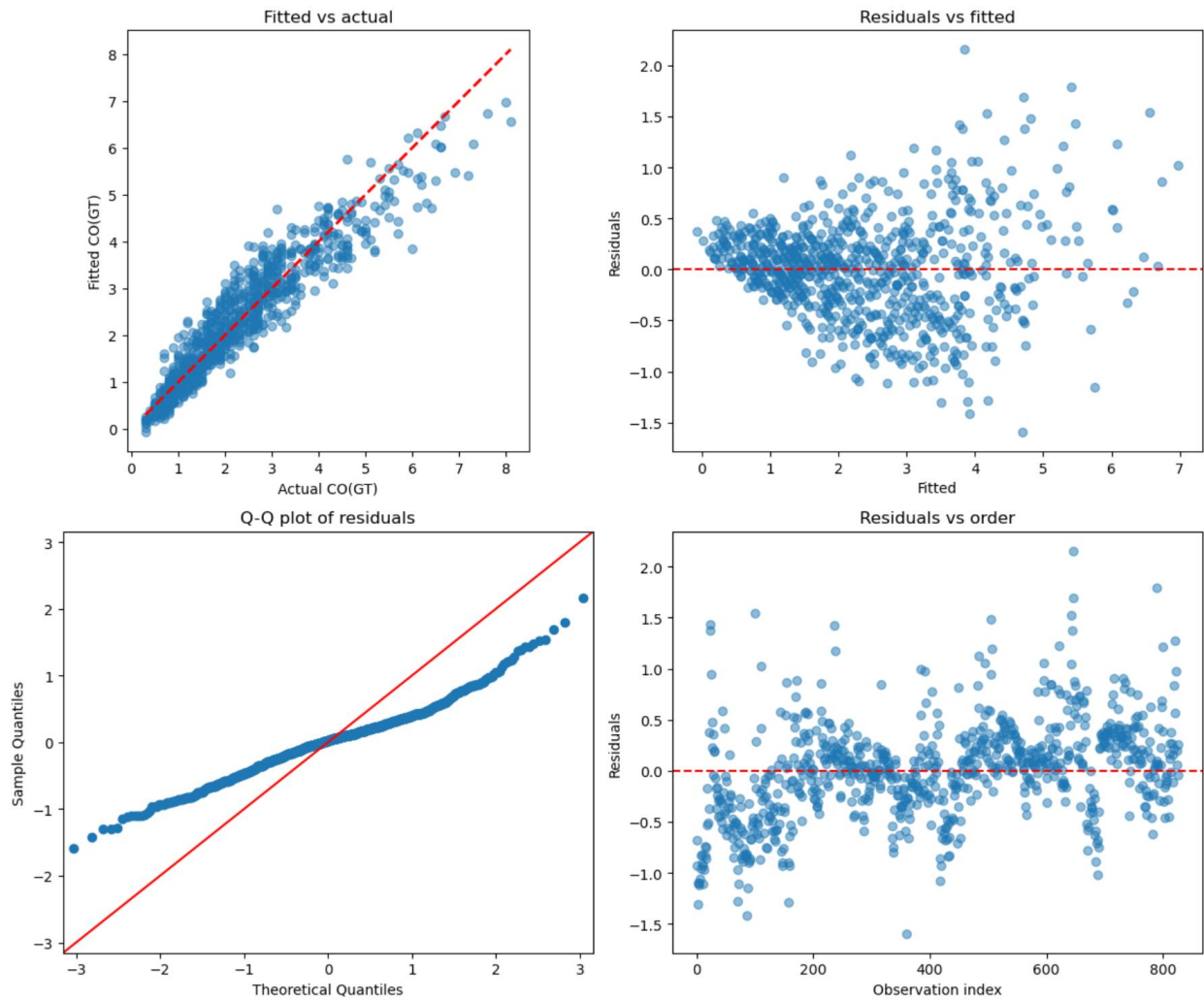
After adding the hour variable, the model was estimated again. The diagnostics in **Figure 10** help check how well the model fits. The fitted vs. actual plot shows that the predictions are

close to the actual values, as most points lie on the diagonal. The residuals vs. fitted plot does not show a clear pattern, suggesting the model is working reasonably well.

Figure 10.

Regression diagnostics for the extended model

(fitted vs actual, residuals, Q-Q plot, residuals over time)



The Q-Q plot shows some differences at the extremes, indicating that the residuals are not perfectly normal, but this is common in real data. The residuals vs. order plot also shows some variation over time, suggesting that the model may still be missing some patterns.

In **Figure 11**, the coefficient plot helps confirm what we were already seeing. The sensor variable remains positive and clearly important, while T and RH maintain a negative relationship with CO(GT). Also, the confidence intervals do not include zero, indicating that these variables are contributing to the model.

Figure 11.

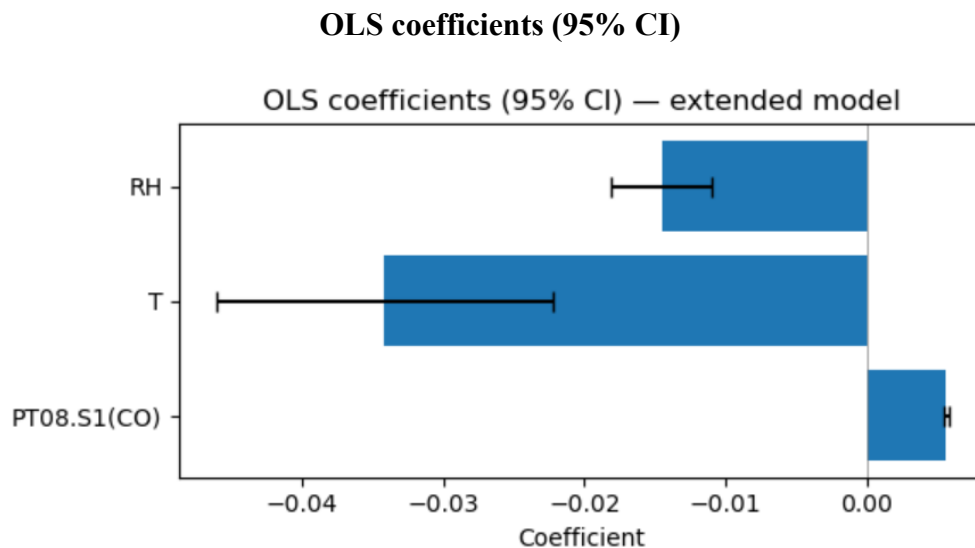
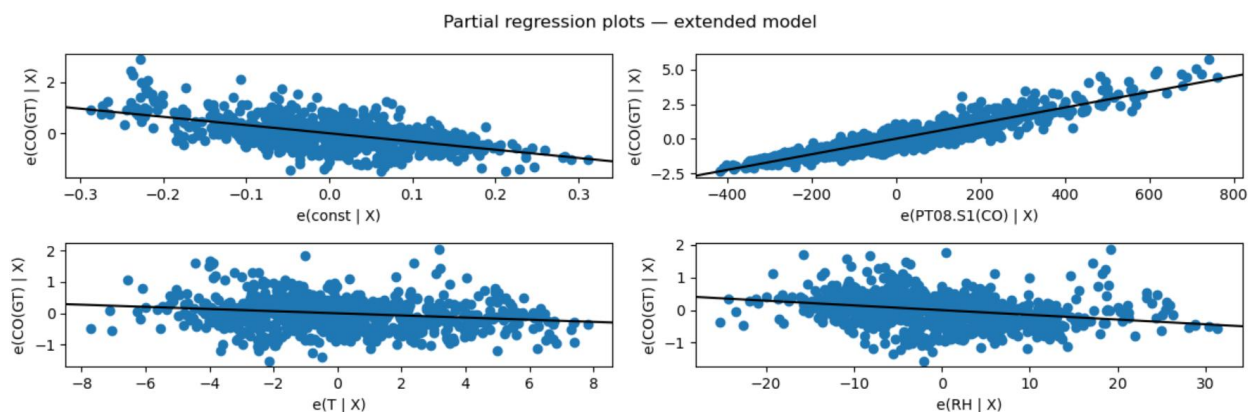


Figure 12 makes this even clearer. The partial plots show how each variable behaves independently. PT08.S1(CO) has a strong and clear relationship with CO, while T and RH have a weaker but consistent negative effect.

Figure 12.

Partial regression plots



Overall, adding the hour variable helps make the model more complete. It does not change the main result, but it adds more context by capturing how CO levels change throughout the day.

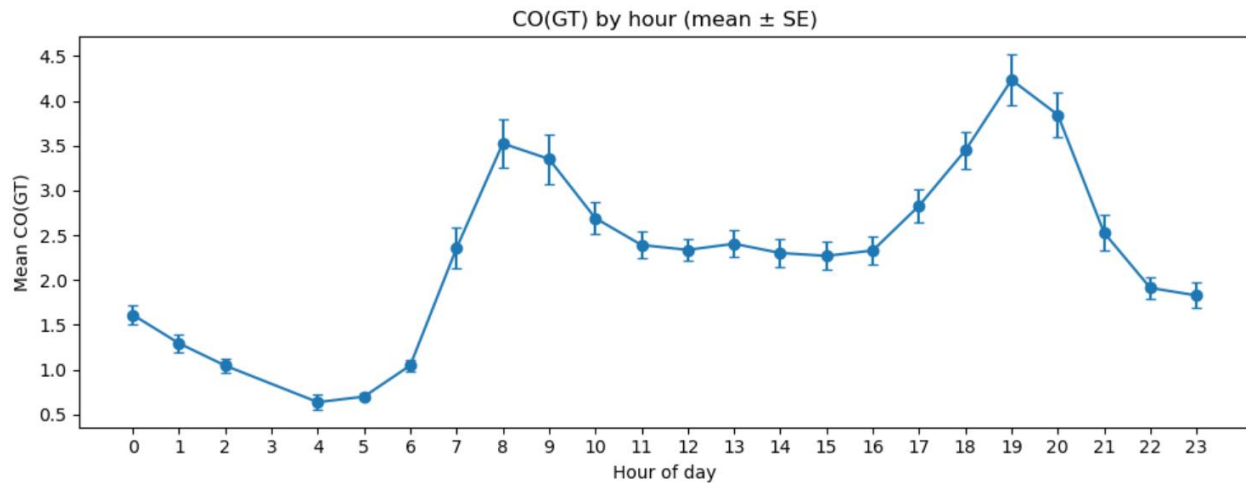
4.1.4. Model 4: Extended with Weekday Effects

To improve the model, time-related variables were added to capture patterns that cannot be explained only by the sensor and weather variables. In this case, both the hour of the day and the day of the week were included. This model represents the most complete specification, as it combines sensor, environmental, and temporal effects into a single framework.

First, the hour variable was created from the time column and added to the regression. As shown in **Figure 13**, CO(GT) changes throughout the day, with higher values in the morning and evening. This suggests that pollution follows a daily pattern, likely related to traffic and daily activity.

Figure 13.

CO(GT) by hour of day (mean $\pm$ SE)



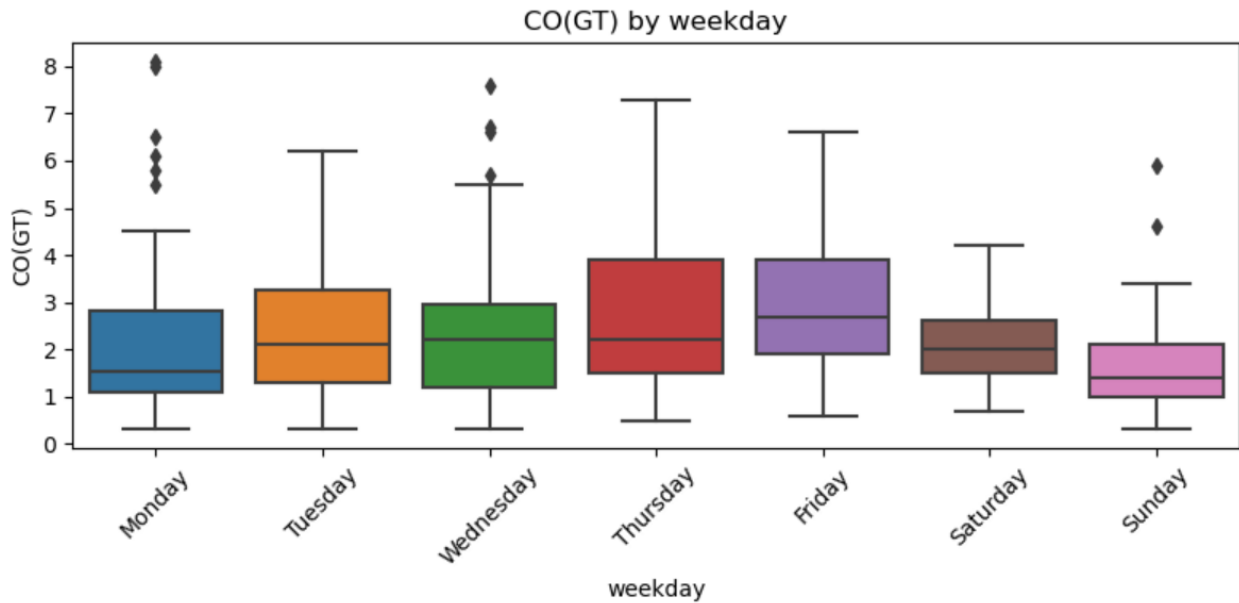
After including hours in the model, its coefficient is positive and statistically significant. This means that the time of day helps explain changes in CO levels. At the same time, the main variables remain consistent: PT08.S1(CO) is still the strongest predictor, while temperature (T) and relative humidity (RH) continue to show a negative relationship with CO(GT).

Next, weekday variables were added using dummy variables. This allows the model to compare each day with a reference category. Based on the results, some days, such as Saturday, Sunday, and Wednesday, show lower CO levels compared to the reference. This may be related to lower activity on weekends or to different patterns during the week.

This is also supported by **Figure 14**, which shows that the distribution of CO(GT) by weekday is slightly lower on weekends.

Figure 14.

Distribution of CO(GT) by weekday



Not all days are significant, indicating that weekday effects exist but are not equally strong on every day.

The regression results in **Figure 15** show that the main variables remain stable. The sensor variable continues to be the strongest predictor, while temperature and humidity keep their negative relationship with CO(GT). The weekday variables do not change the model significantly, and only some of them are statistically meaningful, which suggests that their overall impact is limited.

Figure 15.

OLS Regression Results — Model with Hour and Weekday Effects

OLS Regression Results						
=====						
Dep. Variable:	CO(GT)	R-squared:	0.888			
Model:	OLS	Adj. R-squared:	0.887			
Method:	Least Squares	F-statistic:	1630.			
Date:	Wed, 18 Mar 2026	Prob (F-statistic):	0.00			
Time:	13:47:16	Log-Likelihood:	-551.41			
No. Observations:	827	AIC:	1113.			
Df Residuals:	822	BIC:	1136.			
Df Model:	4					
Covariance Type:	nonrobust					
=====						
	coef	std err	t	P> t	[0.025	0.975]

const	-3.3049	0.160	-20.700	0.000	-3.618	-2.992
PT08.S1(CO)	0.0055	7.98e-05	69.534	0.000	0.005	0.006
T	-0.0342	0.006	-5.682	0.000	-0.046	-0.022
RH	-0.0131	0.002	-7.185	0.000	-0.017	-0.010
hour	0.0111	0.003	4.229	0.000	0.006	0.016
=====						
Omnibus:	27.451	Durbin-Watson:	0.547			
Prob(Omnibus):	0.000	Jarque-Bera (JB):	55.750			
Skew:	0.179	Prob(JB):	7.84e-13			
Kurtosis:	4.221	Cond. No.	1.20e+04			
=====						

Overall, adding time variables improves the model slightly. The R^2 increases from about 0.888 (with hour) to 0.891 (with weekday). The change is small, but it shows that time patterns add extra information to explain CO(GT), even if most of the variation is already captured by the sensor and weather variables.

4.2 Model Diagnostics

The diagnostic plots (**Figures 10–12**) are based on the extended model with hour and are used to evaluate how well the regression is performing.

These results can also be extended to the model with weekday effects, since both models share the same structure and only differ by the inclusion of additional time variables. This is consistent with the small improvement in performance observed earlier (R^2 increasing from 0.888 to 0.891 in **Figure 15**).

The fitted vs. actual plot shows a strong alignment along the diagonal, meaning that the model predictions are close to the observed CO(GT) values. This supports what was previously observed in the regression results, where the model already showed strong explanatory power. The residuals vs. fitted plot does not show a clear pattern, which suggests that the linear relationship identified earlier (**Figures 7–9**) is appropriate and consistent across the data.

The Q-Q plot shows some deviation at the extremes, indicating that the residuals are not perfectly normal. However, this is expected in real-world environmental data and is consistent with the earlier observed variability in the distributions.

The residuals over time show some variation, suggesting that there may still be small time-related patterns not fully captured, even after including the hour variable. This aligns with the patterns observed in **Figure 13**, where CO levels vary throughout the day.

Overall, the diagnostics confirm that the regression approach is appropriate and reinforce the results obtained in the previous models. There are no major issues that would affect the interpretation of the results. This reinforces that adding time variables improves the model slightly, without changing its overall behaviour.

4.3 Additional Time Insights

To better understand how time affects CO(GT), weekday variables were included in the model using dummy variables. This allows each day to be compared against a reference, making it easier to see if pollution levels change during the week.

Looking at the results, most days show a negative effect compared to the reference category. The clearest case is the weekend: Saturday (-0.3153) and Sunday (-0.2470) have the strongest decreases in CO(GT). This means that pollution levels tend to be lower on weekends, even after accounting for the other variables. This makes sense, since there is usually less traffic and overall activity.

During the week, the pattern is still there but less strong. For example, Wednesday (-0.2451) and Tuesday (-0.1879) also show lower CO levels, although the effect is smaller. This suggests that pollution is not constant throughout the week and can vary with daily routines.

Not all days behave the same way. Thursday (-0.0409), for instance, has a very small effect and is not statistically significant. This means that, in practice, it differs little from the reference day.

Overall, adding weekday variables helps reveal a clear pattern: CO levels tend to be lower on weekends and vary slightly across weekdays. Even if these effects are smaller than the main variables, they help explain how pollution changes over time.

4.4 Out-of-Sample Forecasting

To check how the model performs on new data, a time-based split was used. The dataset was first sorted by date and time to keep the correct order. Then, 80% of the data was used to train the model, and the remaining 20% was used for testing. This approach makes the evaluation more realistic, since the model is tested on future data instead of the same data used for training.

This is consistent with the previous results, where this model showed strong explanatory power and stable coefficients.

For this step, the model includes the main variables: PT08.S1(CO), temperature (T), and relative humidity (RH), as this version is stable and easier to interpret.

After training the model, predictions were made on the test set and compared with the actual CO(GT) values.

The performance of the model was evaluated using three metrics:

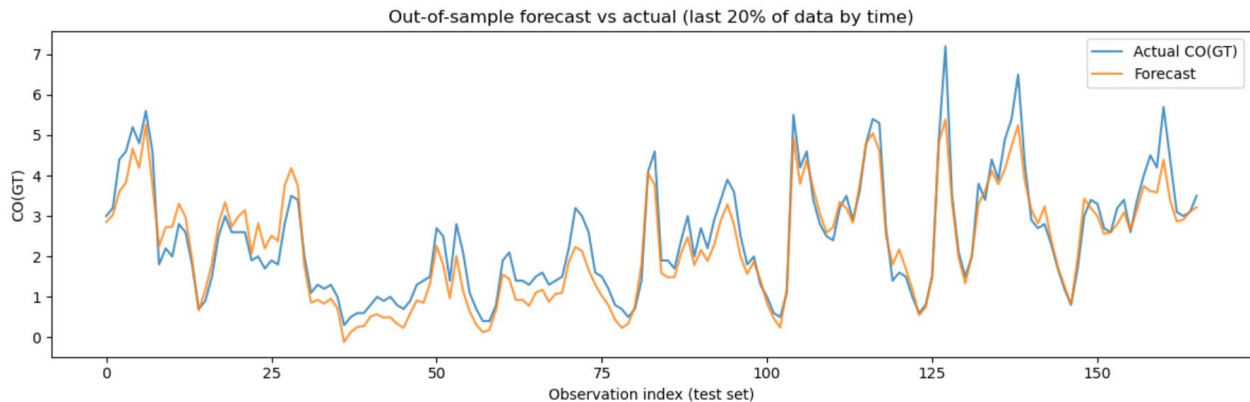
- MAE (Mean Absolute Error) = 0.3874: This represents the average absolute difference between predicted and actual values.
- RMSE (Root Mean Squared Error) = 0.4745: This gives more weight to larger errors, highlighting when the model makes bigger mistakes.
- MAPE (Mean Absolute Percentage Error) = 20.75%: This shows the error as a percentage, making it easier to understand the model's accuracy in relative terms.

Overall, these results indicate that the model makes moderate errors. The difference between MAE and RMSE suggests that there are some larger errors, but they are not too frequent.

As shown in **Figure 16**, the model predictions closely follow the observed CO(GT) values over the test period. Most of the movements are captured, which indicates that the model is working well.

Figure 16.

Out-of-sample prediction vs actual CO(GT) values (test set)



However, the model tends to slightly underestimate some of the highest peaks. This means that while it performs well for general patterns, it is less accurate when CO levels increase sharply.

Overall, the model performs reasonably well on new data. Even though it is not a full forecasting model, it still provides useful and consistent predictions based on the available variables.

5. Evaluation

5.1 Model Performance Metrics

At this evaluation stage, the purpose was to analyze how well the developed models explain and predict carbon monoxide concentration, CO(GT), based on sensor variables, meteorological conditions, and temporal components. In order to go beyond a purely descriptive approach, this evaluation is directly based on the results obtained from the code and also compares the behavior of the different model specifications. The following is a summary of the results obtained for each model:

Model	R ²	MAE	RMSE	Interpretation
Model 2: Refined	0.886	0.3583	0.4758	Stable and easy to interpret
Model 3: With hour	0.888	0.3541	0.4713	Capture better daily pattern
Model 4: With weekday	0.891	0.3583	0.4764	Better fit, small improvement

As a first step, the refined model, built using PT08.S1(CO), temperature (T), and relative humidity (RH), achieved an R² of approximately **0.886**. This indicates that the model explains a very high proportion (**88.6%**) of the observed variability in CO levels. Then, after incorporating the variable hour, the explanatory power increased slightly to **0.888**, suggesting that the time of day provides some additional information about pollution behavior. Finally, the most extended

model, which also includes weekday effects through dummy variables, reached an **R² of 0.891**, which is the highest among the models applied.

As observed, the improvement across models is small; however, it is consistent with the findings from the exploratory analysis and temporal visualizations. In other words, the sensor remains the main driver of the model, but including time-related variables helps capture more systematic patterns in CO variation throughout the day and the week.

On the other hand, an out-of-sample validation was performed using a time-based split: the first 80% of the data was used for training and the remaining **20%** for testing. In this evaluation, the model achieved an **MAE of 0.3874**, an **RMSE of 0.4745**, and a **MAPE of 20.75%**. These results indicate that, on average, the model's absolute error is moderate and that its predictions follow the real evolution of CO(GT) reasonably well in the test set.

Overall, the metrics indicate that the regression approach is appropriate for this study. The model shows high explanatory power within the sample and also an acceptable predictive capacity on unseen observations.

5.2 Performance Summary and Comparative Interpretation

When comparing the developed models, it is observed that the main predictive structure remains stable across all specifications: the variable **PT08.S1(CO)** consistently appears as the most important predictor, which also aligns with the correlation matrix from the exploratory analysis, where it showed a correlation of approximately **0.94** with CO(GT).

The baseline model already showed strong performance but included a significant issue of multicollinearity among meteorological variables, especially T, RH, and AH. This affected the stability of the coefficients and made interpretation more difficult. After removing AH, the model maintained the same level of fit (**R² = 0.886**) and significantly improved in statistical stability.

The VIF values decreased to acceptable levels (See Figure 6), and both T and RH became statistically significant. This result is important because it shows that the refinement of the model was not only a technical simplification but a real improvement in interpretability.

Subsequently, by incorporating the variable hour, the model fit improved slightly. This result is consistent with the CO(GT) by hour plot (See Figure 13), where a clear pattern was observed: CO levels are not constant but tend to increase during certain time periods, likely related to higher urban activity and traffic. Therefore, although the increase in R^2 was small, the inclusion of *hour* adds analytical value by capturing a temporal dimension not reflected solely by sensor and meteorological variables.

Finally, the model with hour + weekday (See figure 14 – 15), achieved the highest overall fit ($R^2 = 0.891$). However, the improvement compared to the previous model was limited. Additionally, the results show that not all weekdays are statistically significant. This suggests that weekly effects exist but are secondary compared to the dominant role of the sensor and, to a lesser extent, meteorological conditions.

In summary, two main conclusions can be drawn. First, if interpretation, stability, and simplicity are prioritized, the refined model without AH represents a very strong option. Second, if a more complete explanation of the temporal behavior of pollution is desired, the model including hourly and weekly variables provides the best fit, although the practical gain is moderate.

5.3 Reliability of Results

In this project, the reliability of the results was evaluated based on three main elements: statistical stability, regression diagnostics, and out-of-sample predictive performance.

Regarding statistical stability, one of the most important findings during the coding process was the detection of multicollinearity in the initial model. The VIF analysis showed that AH presented significant redundancy with other meteorological variables, especially T and RH. This explained why, in the baseline model, several environmental variables were not statistically significant despite having a conceptual relationship with the phenomenon. After removing AH, the VIF values dropped below 5, and both T and RH became significant. This strengthens the reliability of the refined model, as the coefficients become more stable and easier to interpret.

The graphical diagnostics also support the validity of the approach. In the fitted vs. actual plot (**See Figure 10**), most points align closely with the diagonal, indicating that the model predictions follow the observed values well. In the residuals vs. fitted plot (**See Figure 10**), no strong pattern is observed, confirming that the linear form of the model is appropriate. The Q-Q plot (**See Figure 10**), shows deviations at the extremes, indicating that perfect normality of residuals is not fully met. However, this behavior is common in real environmental data, especially when extreme pollution events occur.

The residuals over time plot also shows some variation, suggesting that the model does not fully capture the temporal dynamics of the phenomenon. This is consistent with another finding from the code: in the out-of-sample forecast, the model follows the general trend well but underestimates some of the highest peaks of CO(GT). In other words, the model is reliable for representing overall pollution behavior, but less precise when irregular or extreme events occur.

Additionally, it is important to note that after the data cleaning process, the sample size was reduced from 9,357 observations to 827 complete cases. This decision was methodologically justified, as the value -200 represented faulty or invalid sensor readings. However, this

significant reduction in sample size may limit the variety of observed conditions and should be considered when interpreting the results.

In conclusion, the results can be considered reliable within the studied context: the model is statistically consistent, interpretable, and shows good overall predictive performance. However, its performance decreases for extreme values, and its generalization should be understood within the specific dataset context.

5.4 Final Model Selection

As mentioned previously, the model including hour and weekday variables achieved the highest R^2 value. However, the improvement compared to previous models was relatively small. The increase from 0.886 to 0.891 represents a real improvement, but not large enough to conclude that temporal variables completely transform model performance. On the other hand, the refined model without AH offers a strong combination of simplicity, stability, and interpretability. It eliminates multicollinearity issues and presents a clear structure: the sensor PT08.S1(CO) is the dominant predictor, while T and RH provide additional meaningful context.

For this reason, it can be argued that the refined model is the most appropriate as the main model from a practical and analytical perspective. At the same time, the extended model with hour and weekday can be presented as a complementary version useful for exploring temporal patterns and enriching interpretation.

The selection of the best model should not rely solely on the highest R^2 value, but also on stability, interpretability, and usefulness for decision-making.

5.5. Cross Validation and Model Generalization

In order to understand the reliability of the model a cross-validation approach was incorporate into the study. The time based and train test split can be considered as an starting point, but cross validation offers a more complete view because this approach testing the model across multiple data partitions instead of depending on a single split.

For this study, a k-fold cross-validation was the technique applied, this practice divided the dataset into several subsets, and the model was trained and tested several times using different combination of the folds. This process reduces the risk of results being driven by a particular data split and offer a more stable and trustworthy evaluation. The model was evaluated using the same metrics as before Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and R^2 , in order to ensure consistency in interpretation.

The results obtained indicates that the model performs consistently across the different folds, and there were only minor variations in the metrics used to evaluate. This suggest that the model is sensitive a lot to how the data is split and that the relationships identified especially the strong influence of the PT08.S1(CO) sensor are stable throughout the dataset.

After comparing the results and the findings with the time base out of sample evaluation, the approaches obtained similar conclusions. The model exhibits a good explanatory power and predictive accuracy, although it still struggles slightly with extreme values. The similarity in the results adds confidence to the overall results and supports the idea that the model is valid.

Finally, the incorporation of cross validation strengthens the evaluation process due to this method provide additional evidence that the model is stable and generalizable. It confirms

that the model is not just fitting the data well but is also capable of delivering reliable predictions across different subsets of the dataset.

6. Deployment & Reflection

6.1 Practical Applications

The results of this project show that it is possible to estimate CO(GT) levels using indirect information from sensors and environmental variables, which has practical applications in air quality monitoring.

The strong relationship between PT08.S1(CO) and CO(GT) indicates that this sensor can be used as an early signal of pollution, especially in scenarios where it is not feasible to install specialized equipment across all areas of a city.

Additionally, the inclusion of the *hour* variable confirms that CO levels vary throughout the day, allowing the identification of high-risk periods and improving alert systems. Similarly, weekday effects show that pollution levels tend to be lower on weekends, confirming the influence of human activity.

Overall, these results demonstrate that a simple and well-validated model can support decision-making in environmental monitoring, particularly in smart city strategies and air quality management.

6.2 Limitations

This project presents several relevant limitations. First, the dataset corresponds to the period 2004–2005, representing a specific context that may not reflect current environmental conditions or technology.

Second, the data come from a single city, limiting the applicability of the model to other environments with different climatic conditions or emission sources.

Third, the sample size was significantly reduced to 827 observations after removing invalid values, which implies loss of information and reduced coverage of different scenarios. Additionally, although the model performs well overall, it shows limitations in capturing extreme pollution peaks, as observed in the out-of-sample validation. This suggests the presence of nonlinear relationships or missing factors.

Finally, although temporal evaluation was included, advanced time series models were not applied, which may limit the ability to capture temporal dynamics more accurately.

6.3 Future Improvements

Based on the model results, several improvement opportunities can be identified. First, using more recent data and multiple locations would improve model generalization. Second, advanced models such as Random Forest or Gradient Boosting could better capture nonlinear patterns and improve performance in high pollution peaks.

Additionally, time series techniques such as lag variables or ARIMA/SARIMA models could enhance temporal modeling. Further improvements may include additional features or interactions between sensor, meteorological, and temporal variables.

Finally, future studies could evaluate alternative missing data handling methods, as long as they do not introduce bias.

6.4 Final Insight

This study demonstrates that CO(GT) concentration can be effectively estimated using sensor data, meteorological variables, and temporal factors. The main finding is that PT08.S1(CO) dominates the explanatory power of the model.

Meteorological variables such as T and RH add value only after correcting multicollinearity by removing AH, resulting in a more stable and reliable model. Temporal

variables (hour and weekday) help capture daily and weekly patterns, although their impact on R^2 is moderate.

Overall, the project not only achieves good predictive performance but also demonstrates a solid analytical process aligned with CRISP-DM. It shows that pollution can be modeled using accessible data, provided that appropriate methodological decisions are applied, such as data cleaning, multicollinearity control, and model comparison.

In addition, cross validation was incorporated in order to strengthen the evaluation, this help to confirm that the model performs consistently across different data splits. The results obtained from this approach matched which those obtained from the out of sample testing, the fact reinforce the reliability of the model and its ability to generalize beyond the training data.

The consistency showed adds extra confidence to the findings, it suggest that the relationships identified in the analysis are not driven by a specific partition of the data, and reflect stable and meaningful patterns within the dataset.

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